

Traversing

Traversing is the process of moving an instrument from one station to another along a network of survey lines, also known as traverse legs.

Types of Traverses

Based on Shape:

- **Open Traverse:** Starts from one point and ends at another.
 - Examples: Roads, Railways, Canals, Pipelines, Rivers.
- **Closed Traverse (Loop Traverse):** Starts from one point and ends at the same point.
 - Examples: Plots, Lakes, Area calculations.
- **Link / Connecting Traverse:** Starts from one known point and ends at another known point. It's a type of closed traverse.

Based on Direction

- a) Clockwise Traverse
- b) Anticlockwise Traverse (Commonly used)

Methods of Traversing

To run a traverse, we need:

Length measurement: Chain/Tape

Direction measurement: Compass/Theodolite

Methods based on equipment used

1) Chain Traversing:

- Only chain is used
- Chain angles are measured

2) Free Needle Method (Loose Needle Method):

- Compass is used
- North direction is established independently at every station

- Bearings of lines are measured

3) Fast Needle Method:

- Theodolite with compass is used
- North direction is established only at the first station
- Angles are measured at remaining stations
- Run by theodolite Traverse
- Can use a trough compass, line compass, or tubular compass attached to a theodolite

4) By angles alone:

a) Direct Angles / Included Angles:

- Clockwise angle measured from preceding survey line to following survey line
- Ranges from 0° to 360°
- In clockwise traverses, direct angles are exterior angles
- In anticlockwise traverses, direct angles are interior angles

b) Deflection Angles:

- Measured clockwise or anticlockwise from the extension of the preceding line to the following line
- Clockwise deflection is positive or right deflection
- Anticlockwise deflection is negative or left deflection
- Ranges from 0° to 180°
- Mainly used in Roads and Railways

$\text{Deflection angle} = \text{Included angle} - 180^\circ \text{ (Direct)}$
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Included angle = 200

Deflection angle = $200 - 180 = 20$

Checks in a Traverse

In Closed Traverse:

- Sum of interior angles = $(2n-4) 90^\circ$ or $(n-2) 180^\circ$
- Sum of exterior angles = $(2n+4) 90^\circ$
- Algebraic sum of deflection angles = 360°
- $FB = BB \pm 180^\circ$

In Open Traverse:

- $FB = BB \pm 180^\circ$
- Check lines are run using the measurements

Note: For a regular hexagon, One deflection angle = $360^\circ \div 6 = 60^\circ$

Methods for Plotting a Traverse

1) Directly using Scale and Protractor:

- Least accurate method

2) Independent Coordinate System:

All points are plotted with respect to a common origin

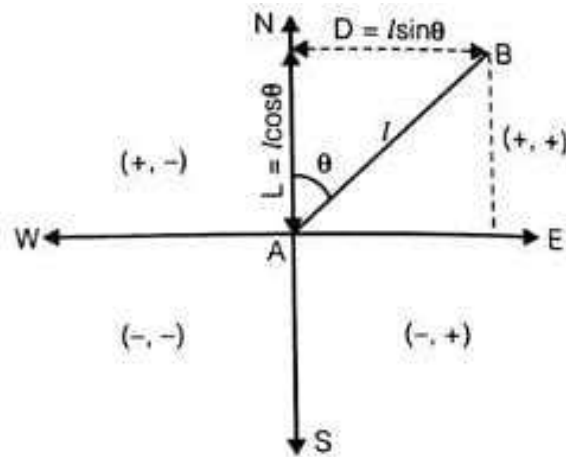
Process: Field measurements → office work → gales traverse table → Traverse computation
→ Coordinates calculation

- Best method for plotting

3) Dependent Coordinate System (Consecutive Coordinate System):

Each point is plotted considering the preceding or previous point as origin

Latitude and Departure



$$\text{Latitude} = l \cos \theta$$

$$\text{Departure} = l \sin \theta$$

In a Closed Traverse:

$$\Sigma \text{Latitude} = 0$$

$$\Sigma \text{Departure} = 0$$

Closing Error

If there exists, traverse will not close when plotted.

$$\Sigma L \neq 0$$

$$\Sigma D \neq 0$$

- Closing error magnitude, $e = \sqrt{(\Sigma L)^2 + (\Sigma D)^2}$
- Direction of closing error, $\theta = \tan^{-1}(\Sigma D / \Sigma L)$
- Relative error of closure = Closing error / Perimeter of traverse

Balancing of Traverse:

The process of applying corrections to latitude and departure of each survey line to make $\Sigma L = 0$ and $\Sigma D = 0$ (to eliminate closing error).

Rules for Applying Corrections:

1. Bowditch's Rule (Compass Rule)

- Applied in compass traverse
- Used when both linear and angular measurements are equally precise.

Assumptions:

1. Error in linear measurement $\propto \sqrt{L}$
2. Error in angular measurement $\propto 1/\sqrt{L}$
3. Errors are accidental in nature

→ Correction for Latitude of a side = $(\Sigma L \times \text{length of the side}) / \text{Perimeter}$ (Same formula for departure)

2. Transit Rule

- Used in theodolite traverse
- Applied when angular measurements are more precise than linear measurements.

→ Correction for latitude of a side = $(\Sigma L \times \text{Latitude of that side}) / \text{Numerical sum of latitudes}$ (Same formula for departure)

Permissible Error in Traverse:

Permissible error = Least Count $\times \sqrt{N}$, where N = Number of Stations

For Theodolite: $e = 20''\sqrt{N}$

For Compass: $e = 15'\sqrt{N}$